Instructor Grading Guide: Lab 1 (Radiometric Correction)

Total Marks: 10 **Course:** RSSI - Spring 2026 **Instructor:** Muhammad Sayed

1. General Assessment Principles

- **Proper Functionality (Crucial):** The output of each task should be as logically expected.
- If a student fails to acheive the desired effect: **up to -100% penalty on that specific task.**
- If the desired outcome is reached: **Full marks eligible.**
- **Vectorization (Crucial):** This lab explicitly forbids `for` loops for pixel processing.
- If a student uses nested loops to iterate over pixels: **-50% penalty on that specific task.**
- If the code is vectorized (uses `numpy`, `scipy.ndimage`): **Full marks eligible.**
- **Reproducibility:** The notebook must run from start to finish without errors. If cells fail to execute due to path errors or missing variables: **-1 Mark overall.**

2. Detailed Rubric (Task by Task)

Task 1: Bad Pixel Correction (3 Marks) *Goal: Remove Salt & Pepper noise using spatial filtering.*

- **[1.0 Mark] Mask Creation:**
- Correctly identifies pixels == 0 (Pepper) and > 9000 (Salt).
- *Partial:* Creates mask but misses one condition (0.5).
- **[1.0 Mark] Convolution Logic:**
- Correctly defines a kernel (e.g., 3x3 averaging) and applies `convolve` (or median filter).
- *Note:* `median_filter` is actually better for salt-and-pepper, but `convolve` (mean) is what was hinted. Accept either.
- **[1.0 Mark] Selective Replacement:**
- Uses `np.where` (or boolean indexing) to replace **only** the bad pixels.
- *Fail:* If they blur the *entire* image (i.e., return the smoothed image directly), give **0** for this part.

Task 2: Line Destriping (2 Marks) *Goal: Remove horizontal sensor bias.*

- **[1.0 Mark] Slicing Logic:**
- Correctly targets the specific rows/bands.
- Ideally uses slicing `[band, row, :]` or `[band, 100, :]`.
- **[1.0 Mark] Correction Math:**
- Subtracts the bias value.

- Includes `np.clip` to prevent underflow (negative values).

Task 3: Haze Removal (3 Marks) *Goal: Dark Object Subtraction (DOS).*

- **[1.0 Mark] Histogram/Percentile Logic:**
- Calculates the 0.1 (or 1st) percentile per band.
- *Fail:* Hardcoding a random number like "500" without calculation.
- **[1.0 Mark] Subtraction:**
- Subtracts this bias from the entire band.
- **[1.0 Mark] Visual/Logic Check:**
- The resulting image histogram starts near 0.
- Negative values are clipped.

Task 4: Evaluation & Metrics (2 Marks) *Goal: Quantitative validation against Ground Truth.*

- **[1.0 Mark] Calculation Implementation:**
- Correctly implements L1 (Mean Abs) or L2 (Mean Squared) logic.
- Data must be normalized (0 to 1) or handled consistently.
- **[1.0 Mark] Accuracy:**
- **Full Mark:** If MSE Loss < 0.01, MAE Loss < 0.1 (on normalized scale).
- **Partail or Zero:** If Loss is higher (indicates poor tuning of noise/haze parameters).

3. Quick Reference Grading Table

Student ID	T1 (Noise) /3	T2 (Stripe) /2	T3 (Haze) /3	T4 (Eval) /2	Penalty	Total /10	Comments
9XXXXXX							